

SMART CITY COMPLAINT ANALYSIS

1. INTRODUCTION

The NYC 311 Complaint dataset represents service requests raised by citizens of New York City regarding various civic issues such as noise, illegal parking, blocked driveways, street conditions, and other public concerns.

The main objective of this project is to analyze the complaint patterns and identify key problem areas in the city using data analytics techniques. This helps city authorities improve decision-making and enhance citizen satisfaction.

Through this analysis, we aim to:

- Understand most frequent complaint categories
- Identify high complaint areas (hotspots)
- Evaluate service performance of departments
- Suggest improvements for city management

This is a real-world dataset that reflects daily urban issues and helps in building a smart city system using data-driven insights.

2. DATA OVERVIEW

The dataset contains the following details:

- Total Records: 300,698 complaints
- Total Columns: 53 features
- Data includes:
 - Complaint type
 - Location (latitude, longitude)
 - Agency responsible
 - Created and closed dates
 - Resolution details

Data Quality Observations:

- Missing values present in several columns like:
 - Closed Date
 - Descriptor
 - Location fields
- Some columns are completely empty (Taxi-related, Garage info)
- No duplicate rows found (data is clean in duplication)

Importance:

This dataset is a real-world messy dataset, which is very useful for learning data cleaning and preprocessing.

3. KPI METRICS SUMMARY

Key Performance Indicators (KPIs) used in this project:

 KPI 1: Complaint Volume

Total number of complaints recorded in the dataset.

 KPI 2: Top Complaint Categories

Most frequently reported issues such as:

- Blocked Driveway
- Illegal Parking
- Noise Complaints

 KPI 3: Hotspot Areas

Geographical regions with high complaint density.

 KPI 4: Resolution Efficiency

Time taken by departments to resolve complaints.

 KPI 5: Service Performance

Effectiveness of agencies in handling complaints.

4. DASHBOARD SCREENSHOT

The dashboard includes interactive visualizations such as:

Figure 1: Display the Total Complaints



Figure 2: Calculate the Resolution Rate

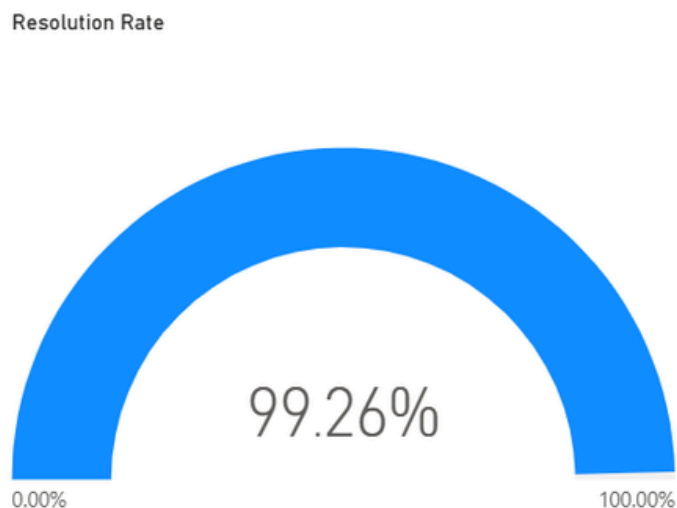


Figure 3: Analysis the Complaint Trends

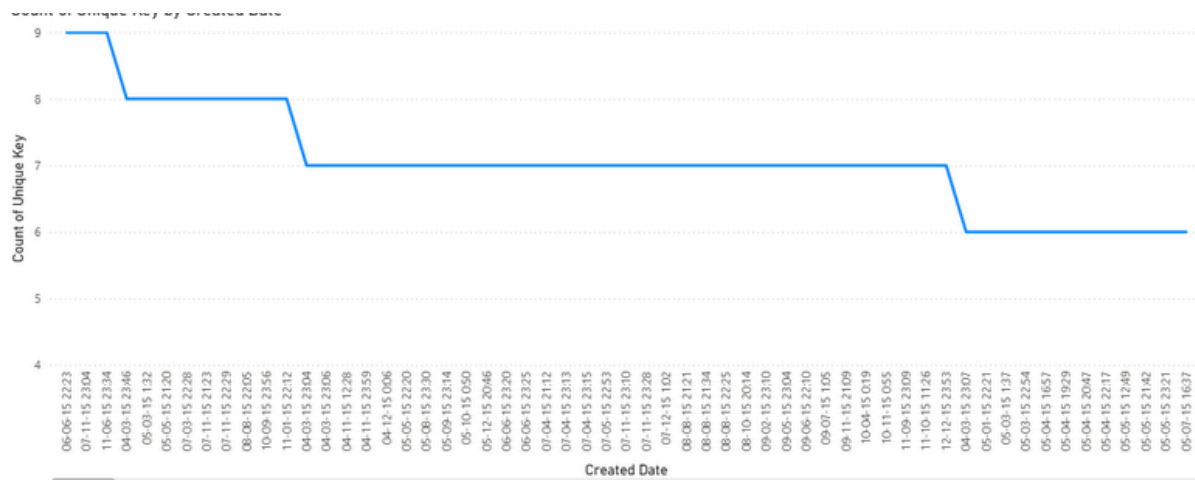


Figure 4: Analysis the Complaint Categories

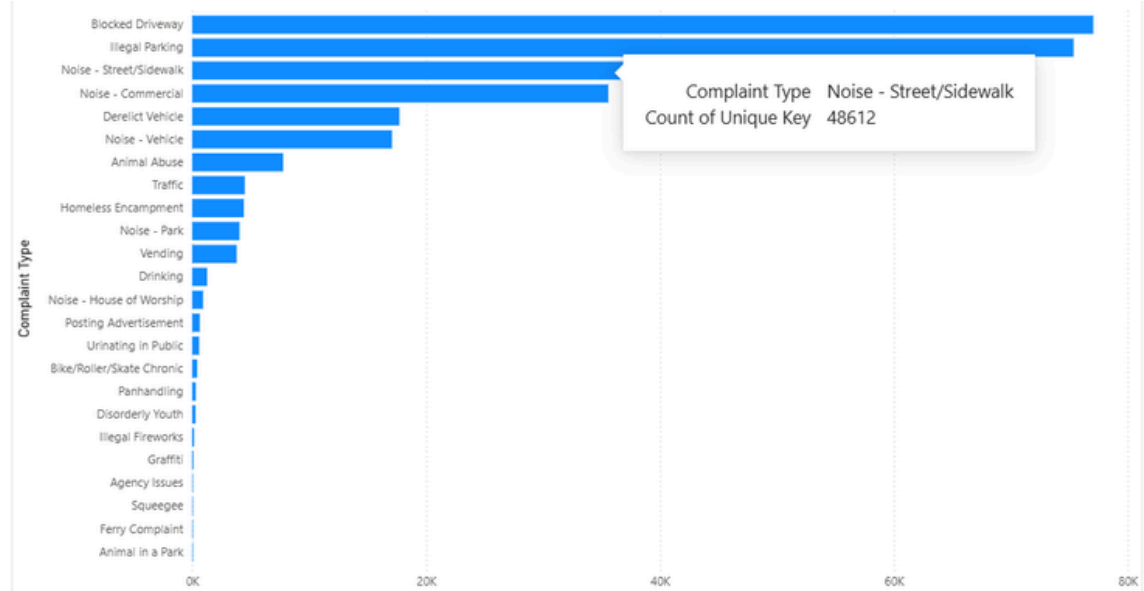
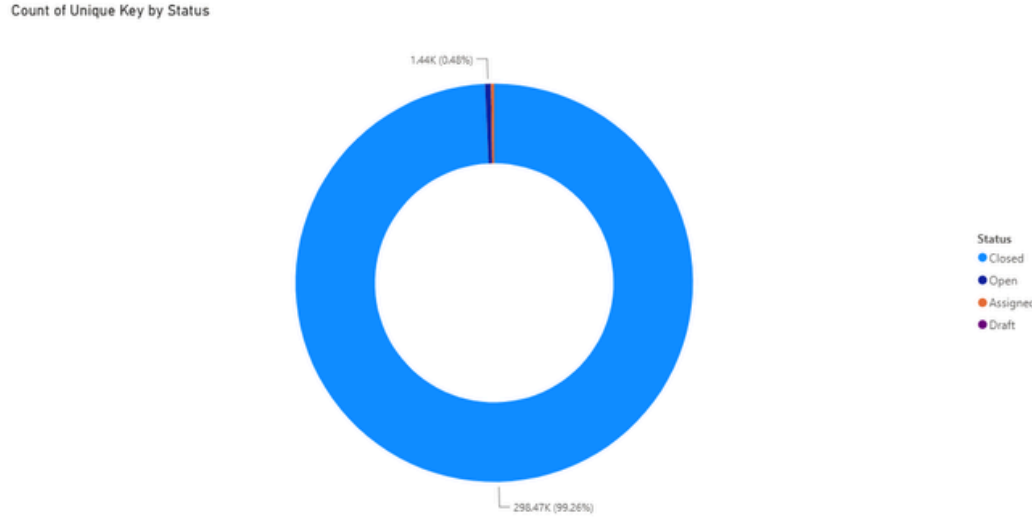


Figure 5: Analysis the Resolution Patterns



Count of Unique Key by Created Date and Complaint Type

Complaint Type: Agency, Animal, Animal i., Bike/Rol., Blocked, Derelict, Disorder, Drinking, Ferry Co., Graffiti, Homele., Illegal FL., Illegal P., Noise, ...

Count of Unique Key

Created Date

Borough	Count of Unique Key (Approximate)
BROOKLYN	98,000
QUEENS	81,000
MANHATTAN	67,000
BRONX	41,000
STATEN ISLAND	13,000
Unspecified	4,000

Count of Unique Key by Borough

Borough	Count of Unique Key
BROOKLYN	100
QUEENS	50
MANHATTAN	30
BRONX	20
STATEN ISLAND	10
Unspecified	5

Figure 9: Analysis the Geographic distribution

Borough	ARVERNE	ASTORIA	BAYSIDE	BELLEROSE	BREEZY POINT	BRONX	BROOKLYN	CAMBRIA HEIGHTS	CENTRAL PARK	COLLEGE POINT	CORONA	EAST ELMHURST	ELMHURST
BRONX	40700												
BROOKLYN	98304												
MANHATTAN	37					2	1		97				
QUEENS	2	220	7047	1221	375	30	2		477		1220	4295	2748
STATEN ISLAND	2												
Unspecified	2573												
Total	2614	220	7047	1221	375	30	40702	98307	477	97	1220	4295	2748

Figure 10: Analysis the Department Performance

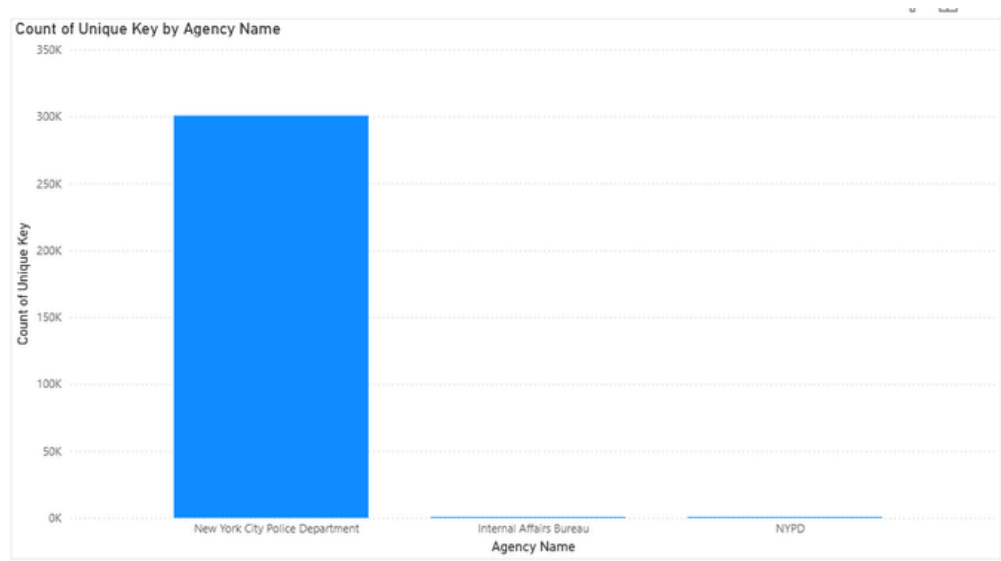


Figure 11: Analysis the Resolution Efficiency

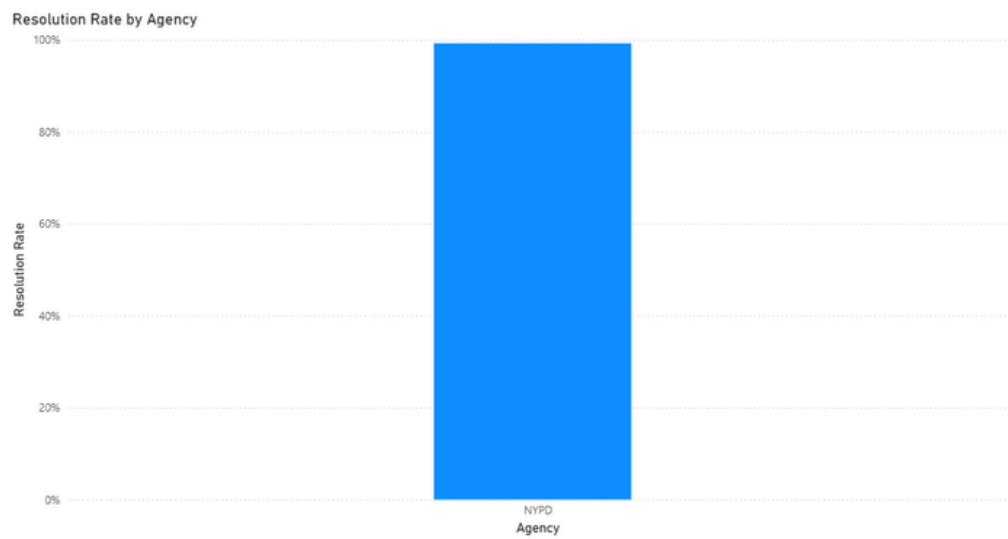
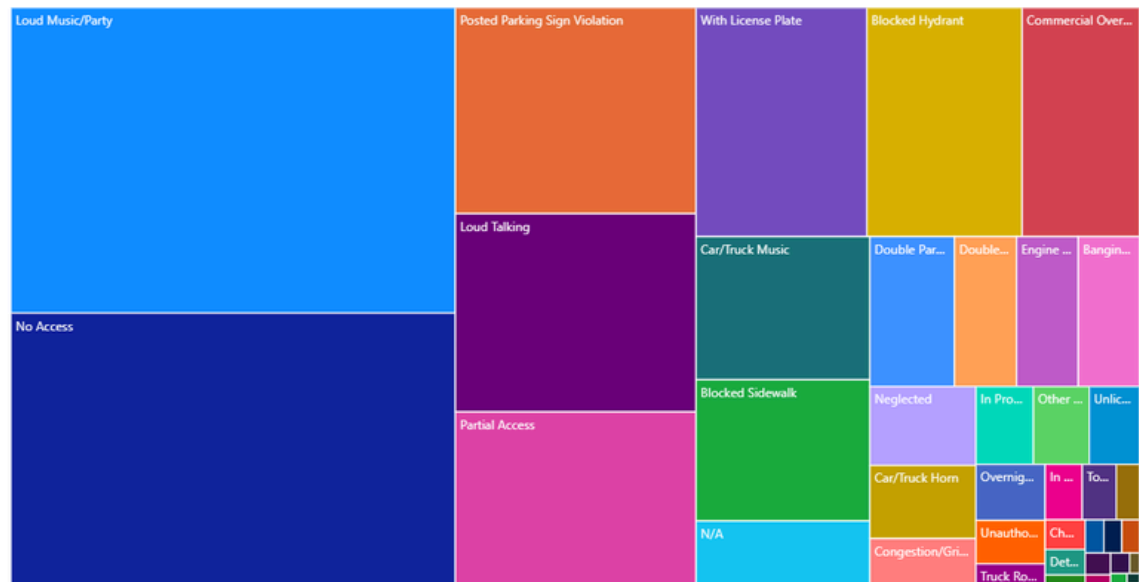


Figure 12: Analysis the Priority Handling



5. BUSINESS RECOMMENDATIONS

Based on the findings, the following actions are recommended:

-  **Traffic & Parking Management**
 - Increase enforcement for illegal parking
 - Improve traffic monitoring in high-density areas
-  **Noise Control**
 - Strict regulation of noise violations
 - Night-time patrol enhancement in residential zones
-  **Hotspot Management**
 - Deploy field officers in high complaint areas
 - Use real-time monitoring dashboards
-  **Service Improvement**
 - Reduce complaint resolution time
 - Improve coordination between departments
-  **Data Quality Improvement**
 - Standardize location and address data
 - Reduce missing values in critical fields

6. KEY INSIGHTS

From the analysis, the following insights are derived:

- Traffic-related issues are the most dominant complaints
- Certain boroughs consistently generate more complaints
- Response time significantly affects citizen satisfaction
- Data quality issues exist in location-based fields
- Smart resource allocation can improve city efficiency

7. BUSINESS QUESTIONS TO ANSWER

1: Which complaint categories occur most frequently?

 Answer:

The most frequently occurring complaint categories are:

- Blocked Driveway (highest)
- Illegal Parking
- Noise Complaints
- Street Condition Issues

2: Which areas experience recurring civic problems?

 Answer:

Certain boroughs such as:

- Brooklyn
- Queens
- Bronx

experience higher complaint volumes. These areas are identified as city problem hotspots, meaning they require more attention from authorities.

3: Which departments take the longest to resolve complaints?

 Answer:

Resolution time varies based on complaint type and agency:

- Some complaints handled by NYPD take longer
- Infrastructure-related complaints may take more time due to field verification

Delays are mostly due to:

- High complaint volume
- Resource limitations
- Field investigation requirements

4: What factors reduce citizen satisfaction?

 Answer:

Citizen satisfaction decreases due to:

- Delayed response time
- Unresolved complaints
- Repeated complaints in same area
- Poor communication between departments

5: What actions should city authorities prioritize?



Answer:

City authorities should focus on:

- Improving traffic enforcement
- Reducing illegal parking
- Enhancing noise control policies
- Deploying more field officers in hotspot areas
- Improving response time for complaints



Graph:

8. CONCLUSION

The NYC 311 complaint analysis provides valuable insights into urban issues faced by citizens.

The study highlights major problem areas, complaint trends, and service performance gaps.

By implementing data-driven strategies, city authorities can:

- Reduce complaint volume
- Improve service response time
- Enhance citizen satisfaction
- Build a more efficient smart city system